Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics 7: $v^2 - u^2 = 2\,a\,s$

One equation that links speeds and distance, with no time in it at all.

Background Knowledge

Read this as you come in. Highlight the key words, and use the space to add your own notes.

A. Distance, speed and velocity

	Scalar — size only	**Vector** — size **and** direction
How far	**distance**, in metres (m)	**displacement**: how far from the start, and which way
How fast	**speed**, in metres per second (m/s)	**velocity**: speed in a stated direction

$\boxed{s = v\,t}$ distance travelled = speed × time

A car at 20 m/s for 5.0 s travels $20 \times 5.0 = 100$ m. This works when the speed is **steady**. When the speed changes, use the **average** speed.

B. Acceleration

Acceleration is the **rate of change of velocity**: how much the velocity changes each second.

$\boxed{a = \dfrac{\Delta v}{t}}$ acceleration = change in velocity ÷ time $\qquad \Delta v = v - u$ (final − initial)

- u is the **initial** velocity, v the **final** velocity. Unit of acceleration: **m/s^2** (metres per second, every second).
- Speeding up: a is **positive**. Slowing down: a is **negative**. Finish minus start, in that order, and the sign comes out right by itself.
- **Uniform** acceleration means the velocity changes by the same amount every second. An object falling freely near the Earth accelerates at about **9.8 m/s^2**.

A bike goes from 2.0 m/s to 8.0 m/s in 3.0 s: $a = (8.0 - 2.0) \div 3.0 = 2.0$ m/s^2.

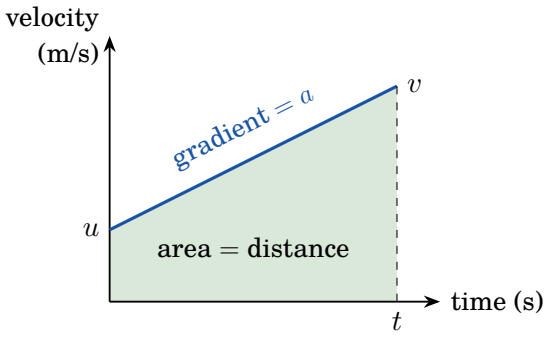
C. Reading the motion graphs

Distance–time graph

- **gradient = speed**
- a flat line: stationary
- a steeper line: faster

Velocity–time graph

- **gradient = acceleration**
- **area under the line = distance travelled**
- a flat line: constant velocity



$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

A straight sloping line on a velocity–time graph means **uniform** acceleration. That is the case today's equation is built for.

D. How this ties together — and where today goes

Everything so far has **time** in it: $s = v\,t$, and $a = \Delta v \div t$.

Plenty of real questions give you **no time at all**. A car brakes from 20 m/s to a stop: how far does it skid? An aeroplane needs to reach 60 m/s: how long must the runway be? You know the speeds and the acceleration, and you want the distance.

Two equations you already have, written with u and v:

① $s = \frac{1}{2}(u + v)\,t$ — average velocity × time: the area under the graph in C

② $v = u + a\,t$ — acceleration from B, rearranged

Both have t in them. Put them together and the t drops out, leaving one equation with no time in it:

$$\boxed{v^2 - u^2 = 2\,a\,s}$$

Today's question: if you know how fast something starts, how fast it finishes and how hard it accelerates, how far does it travel?

The equation — it is on your equation sheet

$$\boxed{v^2 - u^2 = 2\,a\,s}$$

(final velocity)^2 − (initial velocity)^2 = 2 × acceleration × distance

Symbol	Quantity	Unit
s	distance	m
u	initial velocity	m/s
v	final velocity	m/s
a	acceleration	m/s^2
t	time	s

The Givens — start every calculation with this box

Givens

$s =$
$u =$
$v =$
$a =$
$t =$

Write the five letters **down the page**, always in this order: s, u, v, a, t. Read downwards, they spell **SUVAT**.

- Fill in each one the question gives you, with its unit.
- Put **?** next to the one you want.
- Leave the others empty.

The **empty** line is the quantity the question leaves out. When t is empty, use $v^2 - u^2 = 2\,a\,s$: the equation with **no *t***.

Worked example 1 — find the distance

A cyclist speeds up from 3.0 m/s to 7.0 m/s with an acceleration of 0.50 m/s^2. Calculate the distance travelled while accelerating.

1. Givens

$s =$ **?**
$u = 3.0$ m/s
$v = 7.0$ m/s
$a = 0.50$ m/s^2
$t =$

2. Equation $\quad v^2 - u^2 = 2\,a\,s$

3. Substitute $\quad 7.0^2 - 3.0^2 = 2 \times 0.50 \times s$

4. Rearrange

$$s = \frac{7.0^2 - 3.0^2}{2 \times 0.50}$$

5. Calculate

$$s = \frac{49 - 9.0}{1.0} = \mathbf{40\ m}$$

Worked example 2 — find the final velocity

A trolley moving at 5.0 m/s accelerates at 2.0 m/s^2 over a distance of 30 m. Calculate its final velocity. Give your answer to 2 significant figures.

1. Givens

$s = 30$ m
$u = 5.0$ m/s
$v =$ **?**
$a = 2.0$ m/s^2
$t =$

2. Equation $\quad v^2 - u^2 = 2\,a\,s$

3. Substitute $\quad v^2 - 5.0^2 = 2 \times 2.0 \times 30$

4. Rearrange

$$v^2 = 5.0^2 + 2 \times 2.0 \times 30$$
$$v = \sqrt{5.0^2 + 2 \times 2.0 \times 30}$$

5. Calculate

$$v = \sqrt{25 + 120} = \sqrt{145}$$
$$= 12.04\ldots = \mathbf{12\ m/s}\ \text{(2 s.f.)}$$

The check. Put your answer back into $v^2 - u^2 = 2\,a\,s$ and work out **each side on its own**. The two sides give the **same number**. In worked example 1: $v^2 - u^2 = 49 - 9 = 40$ and $2\,a\,s = 2 \times 0.50 \times 40 = 40$. They match, so $s = 40$ m is right.

Questions

Fill in the Givens box first, then follow the worked examples: equation, substitute under each letter, rearrange, calculate, and give the answer with its unit. Use the check on at least one answer in each section.

Everyone — *find the distance*

1) **Calculate** the distance a runner covers while accelerating from rest to $8.0\,\text{m/s}$ at $2.0\,\text{m/s}^2$. **[3]**

Givens

$s =$ **?**
$u = 0$
$v = 8.0\ \text{m/s}$
$a = 2.0\ \text{m/s}^2$
$t =$

$$v^2 - u^2 = 2 \quad a \quad s$$
$$8.0^2 - 0^2 = 2 \times 2.0 \times s$$
$$s = \frac{8.0^2 - 0^2}{2 \times 2.0} = \frac{64}{4.0} = \mathbf{16\ m}$$

2) **Calculate** the distance a car travels while it speeds up from $6.0\,\text{m/s}$ to $14\,\text{m/s}$ with an acceleration of $1.6\,\text{m/s}^2$. **[3]**

Givens

$s =$ **?**
$u = 6.0\ \text{m/s}$
$v = 14\ \text{m/s}$
$a = 1.6\ \text{m/s}^2$
$t =$

$$v^2 - u^2 = 2 \quad a \quad s$$
$$14^2 - 6.0^2 = 2 \times 1.6 \times s$$
$$s = \frac{14^2 - 6.0^2}{2 \times 1.6} = \frac{160}{3.2} = \mathbf{50\ m}$$

3) **Calculate** the distance a skateboarder rolls downhill while speeding up from $1.0\,\text{m/s}$ to $5.0\,\text{m/s}$ with an acceleration of $0.80\,\text{m/s}^2$. **[3]**

Givens

$s =$ **?**
$u = 1.0\ \text{m/s}$
$v = 5.0\ \text{m/s}$
$a = 0.80\ \text{m/s}^2$
$t =$

$$v^2 - u^2 = 2 \quad a \quad s$$
$$5.0^2 - 1.0^2 = 2 \times 0.80 \times s$$
$$s = \frac{5.0^2 - 1.0^2}{2 \times 0.80} = \frac{24}{1.6} = \mathbf{15\ m}$$

4) **Calculate** the braking distance of a car that slows from $20\,\text{m/s}$ to a stop with an acceleration of $-5.0\,\text{m/s}^2$. A distance comes out positive. **[3]**

Givens

$s =$ **?**
$u = 20\ \text{m/s}$
$v = 0$
$a = -5.0\ \text{m/s}^2$
$t =$

$$v^2 - u^2 = 2 \quad a \quad s$$
$$0^2 - 20^2 = 2 \times (-5.0) \times s$$
$$s = \frac{0^2 - 20^2}{2 \times (-5.0)} = \frac{-400}{-10} = \mathbf{40\ m}$$

5) **Calculate** the acceleration of an aeroplane that starts from rest and reaches $60\,\text{m/s}$ along a runway $900\,\text{m}$ long. **[3]**

Givens

$s = 900\ \text{m}$
$u = 0$
$v = 60\ \text{m/s}$
$a =$ **?**
$t =$

$$v^2 - u^2 = 2 \quad a \quad s$$
$$60^2 - 0^2 = 2 \times a \times 900$$
$$a = \frac{60^2 - 0^2}{2 \times 900} = \frac{3600}{1800} = \mathbf{2.0\ m/s^2}$$

6) **Calculate** the final velocity of a ball that is rolling at $3.0\,\text{m/s}$ and then accelerates at $1.5\,\text{m/s}^2$ over $12\,\text{m}$. Give your answer to 2 significant figures. **[3]**

Givens

$s = 12\ \text{m}$
$u = 3.0\ \text{m/s}$
$v =$ **?**
$a = 1.5\ \text{m/s}^2$
$t =$

$$v^2 - u^2 = 2 \quad a \quad s$$
$$v^2 - 3.0^2 = 2 \times 1.5 \times 12$$
$$v = \sqrt{3.0^2 + 2 \times 1.5 \times 12} = \sqrt{45}$$
$$= 6.708\ldots = \mathbf{6.7\ m/s}$$

Stretch — *two equations in one question*

7) A stone is dropped from rest from a bridge and falls $20\,\text{m}$ to the water. It falls with an acceleration of $9.8\,\text{m/s}^2$.

(a) Calculate the speed of the stone as it reaches the water. Give your answer to 2 significant figures.

Givens

$s = 20\ \text{m}$
$u = 0$
$v = \textbf{?}$ (a)
$a = 9.8\ \text{m/s}^2$
$t = \textbf{?}$ (b)

$$v^2 - u^2 = 2 \quad a \quad s$$
$$v^2 - 0^2 = 2 \times 9.8 \times 20$$
$$v = \sqrt{0^2 + 2 \times 9.8 \times 20} = \sqrt{392}$$
$$= 19.80\ldots = \textbf{20 m/s} \textit{ (2 s.f.)}$$

[3]

(b) Use your answer to (a) and $a = \dfrac{\Delta v}{t}$ to calculate how long the stone takes to fall.

$$t = \frac{\Delta v}{a} = \frac{19.8 - 0}{9.8} = 2.02\ldots = \textbf{2.0 s}$$

[2]

8) A train speeds up steadily from $8.0\,\text{m/s}$ to $16\,\text{m/s}$ over a distance of $240\,\text{m}$.

(a) Calculate the acceleration of the train.

Givens

$s = 240\ \text{m}$
$u = 8.0\ \text{m/s}$
$v = 16\ \text{m/s}$
$a = \textbf{?}$ (a)
$t = \textbf{?}$ (b)

$$v^2 - u^2 = 2 \quad a \quad s$$
$$16^2 - 8.0^2 = 2 \times a \times 240$$
$$a = \frac{16^2 - 8.0^2}{2 \times 240} = \frac{192}{480} = \textbf{0.40 m/s}^2$$

[3]

(b) Calculate the time the train takes to cover the $240\,\text{m}$.

$$t = \frac{\Delta v}{a} = \frac{16 - 8.0}{0.40} = \textbf{20 s}$$

[2]

(c) Calculate the average speed of the train over the $240\,\text{m}$. Your answer sits between its initial and final velocities.

$\dfrac{240}{20} = \textbf{12 m/s}$ — *between 8.0 and 16, and exactly halfway, as it is for uniform acceleration.*

[1]